DGMS Examination 2019 — Question 46 Detailed Solution

Junior Scientific Officer (Geology) | Geomorphology, Sedimentology & Geochemistry

QUESTION 46

Which one of the following pairs regarding weathering is NOT correctly matched?

(a) Hydration : Occurs when minerals absorb water molecules

(b) Oxidation : Occurs when an atom or an ion gains an electron
NOT CORRECTLY MATCHED (CORRECT OPTION)

(c) Carbonation : Occurs when carbonates are formed

(d) Chelation : Occurs when metal ion is removed by organic matter

1. Executive Summary & Answer Verification

The correct option to choose is **(b) Oxidation : Occurs when an atom or an ion gains an electron**, because this statement provides an inverted, erroneous scientific definition.

In fundamental chemistry and geochemistry, the mnemonic **OIL RIG** governs redox reactions:

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OIL = Oxidation Is Loss of electrons
RIG = Reduction Is Gain of electrons
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Oxidation is defined as the *loss of electrons* by an atom or ion, which causes an algebraic increase in its positive oxidation state (for example, ferrous iron oxidizing to ferric iron: $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + e^{-}$). In contrast, gaining an electron is the universal definition of **Reduction** ($\text{Fe}^{3+} + e^{-} \rightarrow \text{Fe}^{2+}$). Hence, pair (b) erroneously defines oxidation as electron gain and is NOT correctly matched.

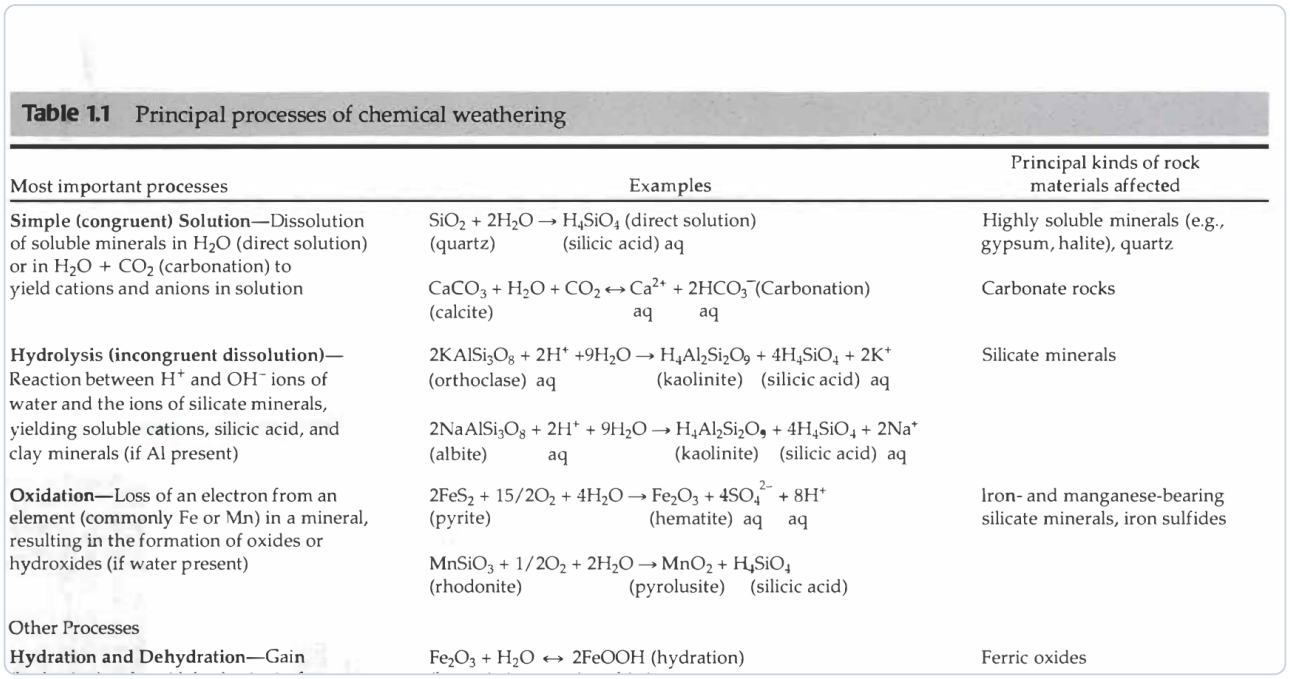


Table 1.1 Principal processes of chemical weathering

Most important processes	Examples	Principal kinds of rock materials affected
Simple (congruent) Solution—Dissolution of soluble minerals in H_2O (direct solution) or in $H_2O + CO_2$ (carbonation) to yield cations and anions in solution	$SiO_2 + 2H_2O \rightarrow H_4SiO_4$ (direct solution) (quartz) (silicic acid) aq	Highly soluble minerals (e.g., gypsum, halite), quartz
	$CaCO_3 + H_2O + CO_2 \leftrightarrow Ca^{2+} + 2HCO_3^-$ (Carbonation) (calcite) aq aq	Carbonate rocks
Hydrolysis (incongruent dissolution)—Reaction between H^+ and OH^- ions of water and the ions of silicate minerals, yielding soluble cations, silicic acid, and clay minerals (if Al present)	$2KAlSi_3O_8 + 2H^+ + 9H_2O \rightarrow H_4Al_2Si_2O_9 + 4H_4SiO_4 + 2K^+$ (orthoclase) aq (kaolinite) (silicic acid) aq	Silicate minerals
	$2NaAlSi_3O_8 + 2H^+ + 9H_2O \rightarrow H_4Al_2Si_2O_9 + 4H_4SiO_4 + 2Na^+$ (albite) aq (kaolinite) (silicic acid) aq	
Oxidation—Loss of an electron from an element (commonly Fe or Mn) in a mineral, resulting in the formation of oxides or hydroxides (if water present)	$2FeS_2 + 15/2O_2 + 4H_2O \rightarrow Fe_2O_3 + 4SO_4^{2-} + 8H^+$ (pyrite) (hematite) aq aq	Iron- and manganese-bearing silicate minerals, iron sulfides
	$MnSiO_3 + 1/2O_2 + 2H_2O \rightarrow MnO_2 + H_4SiO_4$ (rhodonite) (pyrolusite) (silicic acid)	
Other Processes		
Hydration and Dehydration—Gain	$Fe_2O_3 + H_2O \leftrightarrow 2FeOOH$ (hydration)	Ferric oxides

Figure 46.1: *Principal processes of chemical weathering tabulated with definitions, reactions, and affected rock types. Note explicit definition: "Oxidation — Loss of an electron from an element (commonly Fe or Mn) in a mineral..." (Sam Boggs Jr., Principles of Sedimentology and Stratigraphy, 4th ed., Table 1.1, p. 23).*

2. Comprehensive Analysis of Weathering Mechanisms

2.1 Oxidation and Reduction in Chemical Weathering

Oxidation occurs predominantly in subaerial environments exposed to atmospheric oxygen (O_2) and aerated groundwaters. The most abundant transition elements subject to weathering oxidation in common rock-forming minerals are iron (Fe) and manganese (Mn):

- **Iron Oxidation:** Ferrous silicate minerals (e.g., olivine, pyroxene, amphibole, biotite) contain divalent Fe^{2+}. Upon chemical weathering, each Fe^{2+} ion loses one electron:

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\text{Fe}^{2+} \longrightarrow \text{Fe}^{3+} + e^{-}    (\text{Oxidation: loss of electron})
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  Because Fe^{3+} has a smaller ionic radius and higher charge than Fe^{2+}, this charge disturbance destabilizes the parent crystal lattice, promoting disintegration and precipitating insoluble ferric oxyhydroxides such as goethite ($\alpha\text{-FeOOH}$) or hematite (Fe_2O_3):

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4\text{FeSiO}_3 (\text{pyroxene}) + \text{O}_2 + 2\text{H}_2\text{O} \longrightarrow 4\text{FeOOH} (\text{goethite}) + 4\text{SiO}_2 (\text{aq})
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- **Pyrite Oxidation:** In sulfide ores and black shales, pyrite (FeS_2) undergoes intense oxidative weathering, producing sulfuric acid and iron oxides:

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2\text{FeS}_2 + \frac{15}{2}\text{O}_2 + 4\text{H}_2\text{O} \longrightarrow \text{Fe}_2\text{O}_3 + 4\text{SO}_4^{2-} + 8\text{H}^{+}
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2.2 Hydration and Dehydration

Hydration involves the direct incorporation of neutral water (H_2O) molecules into the crystal lattice of an existing mineral without causing comprehensive structural dissolution, thereby forming a hydrated mineral phase with increased molar volume:

- **Anhydrite to Gypsum:** $\text{CaSO}_4 + 2\text{H}_2\text{O} \longrightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (accompanied by ~60% volumetric expansion, exerting significant physical stress).
- **Hematite to Goethite:** $\text{Fe}_2\text{O}_3 + \text{H}_2\text{O} \longrightarrow 2\text{FeOOH}$.

Pair (a) ("Hydration : Occurs when minerals absorb water molecules") is scientifically accurate.

2.3 Carbonation

Carbonation refers to the reaction of minerals with carbonic acid (H_2CO_3), which forms naturally when atmospheric or soil-gas carbon dioxide dissolves in water:

$$\text{CO}_2(\text{g}) + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^{+} + \text{HCO}_3^{-}$$

During carbonation, bicarbonate (HCO_3^{-}) and carbonate salts or aqueous carbonate species are formed. In carbonate terrains:

$$\text{CaCO}_3 (\text{calcite}) + \text{H}_2\text{O} + \text{CO}_2 \rightleftharpoons \text{Ca}^{2+} + 2\text{HCO}_3^{-} (\text{calcium bicarbonate})$$

When dissolved weathering solutions evaporate or lose CO_2, secondary carbonates (such as caliche, speleothems, or travertines) precipitate. Pair (c) is therefore correctly matched.

2.4 Chelation (Organic Complexation)

Chelation (from Greek *chele*, meaning claw) is a biochemical weathering process in which organic molecules with polydentate ring structures (secreted by lichens, mosses, soil fungi, and plant root systems, such as humic, fulvic, and citric acids) bind and "cage" polyvalent metallic cations (particularly Fe^{3+}, Al^{3+}, Mg^{2+}):

$$\text{Metal ion (e.g., Al}^{3+}\text{, Fe}^{3+}\text{) + Organic ligand} \longrightarrow [\text{Organo-metallic Chelate Complex}] + \text{H}^{+}$$

This mobilization dissolves otherwise highly insoluble minerals (such as feldspars, pyroxenes, and clays) and leaches metals downward through soil horizons. Pair (d) ("Chelation : Occurs when metal ion is removed by organic matter") is scientifically accurate.

3. Systematic Comparison of Weathering Processes

Weathering Process	Primary Chemical Agent	Atomic / Molecular Mechanism	Typical Mineral Transformation	Match Status in Question
Hydration	H_2O molecules			Correctly matched

Weathering Process	Primary Chemical Agent	Atomic / Molecular Mechanism	Typical Mineral Transformation	Match Status in Question
		Adsorption / inclusion of water into lattice	Anhydrite $\rightarrow$ Gypsum; Hematite $ \rightarrow$ Goethite	
Oxidation	O_2, $ \text{Fe}^{3+}$, oxidants	**Loss of electrons** (increase in positive valence)	Fe^{2+} (silicates) $\rightarrow \text{Fe}^{3+}$ (oxides/hydroxides)	**NOT correctly matched (Erroneously stated as electron gain)**
Reduction	H_2S, organic carbon, anaerobic water	**Gain of electrons** (decrease in positive valence)	Fe^{3+} oxides $ \rightarrow \text{Fe}^{2+}$ soluble ions in bogs/ swamps	N/A (True definition of electron gain)
Carbonation	$\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H} _2\text{CO}_3$	Proton exchange producing carbonates/ bicarbonates	Calcite dissolution; feldspar weathering yielding $ \text{HCO}_3^{-}$	Correctly matched
Chelation	Organic acids / ligands (fulvic, humic)	Ring-bonding and extraction of metal cations	Extraction of $\text{Al} ^{3+}, \text{Fe}^{3+}$ by lichen acids from basalt	Correctly matched

Textbook Authority

Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (Chapter 1, p. 23–25):
"Oxidation — Loss of an electron from an element (commonly Fe or Mn) in a mineral, resulting in the formation of oxides or hydroxides... An electron is lost from iron during oxidation processes."
William M. White, *Geochemistry* (Chapter 13: Weathering, Soils, and Stream Chemistry):
"Oxidation involves the loss of electrons, while reduction involves the gain of electrons. Because electron transfer must balance, oxidation of one species is always accompanied by reduction of another."